

Foundation (Jalapeno)

- Q1.** What is the x -coordinate of the intersection point of the lines $y = 2x + 1$ and $y = x - 2$?
(A) -3
(B) -1
(C) 1
(D) 3
(E) 5
- Q2.** A tennis player hits a ball at a distance of 15 meters from the net. The ball travels at a rate of 2 meters per second. If the opponent is 20 meters from the net, how many seconds t will it take for the ball to reach the opponent's position, given by $d = 2t + 15$ and $d = t + 20$?
(A) 5
(B) 10
(C) 15
(D) 20
(E) 25
- Q3.** The system of equations $y = x + 2$ and $y = -x - 1$ has a solution at $x =$?
(A) -1
(B) -0.5
(C) 0
(D) 0.5
(E) 1
- Q4.** A basketball team scored x points in the first quarter and y points in the second quarter. The team's total score after two quarters is given by $x + y = 30$. If $x = 2y - 5$, what is y ?
(A) 5
(B) 10
(C) 15
(D) 20
- (E) 25
- Q5.** A swimmer swims at a rate of 1.5 meters per second. If the swimmer starts at one end of a 25-meter pool, how many seconds t will it take to reach the other end, given by $d = 1.5t$ and $d = 25$?
(A) 10
(B) 15
(C) 16.67
(D) 20
(E) 25
- Q6.** The system of equations $y = x - 1$ and $y = -2x + 3$ has a solution at $(x, y) =$?
(A) (1, 0)
(B) (1, 2)
(C) (2, 1)
(D) (2, 3)
(E) (4, 1)
- Q7.** A runner runs x miles in the morning and y miles in the evening. The total distance run is given by $x + y = 10$. If $y = 2x - 2$, what is x ?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q8.** The system of equations $y = x + 3$ and $y = -x - 2$ has a solution at $(x, y) =$?
(A) (-1, 2)
(B) (-2, 1)
(C) (1, 4)
(D) (2, 5)
(E) (0, 1)

Open Ended Questions (FRQ)

- Q9.** A cyclist travels x miles in the first hour and y miles in the second hour. The total distance traveled is given by $x + y = 25$. If $y = 3x - 5$, find the values of x and y and graph the system of equations.
- Q10.** A gymnast scores x points for a vault and y points for a floor exercise. The total score is given by $x + y = 20$. If $x = 2y - 4$, find the values of x and y and interpret the solution graphically.

Chef's Tips

- Use graph paper for graphical solutions
- Label all axes and lines clearly
- Check for consistency between graphical and algebraic solutions